The 3rd Read Head

The Closure Prices the Map Before Any Tour Is Walked

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All load-bearing numbers quoted from live computation this session, on the TSPLIB benchmark set with known optima and, where available, known optimal tours. Two classical constructions are used as beams and marked as such. Mixed results are reported mixed. One aggregate is stated with its non-significant sign test attached.

Abstract

The rendering program ended its last report at a wall: at N = 100–150 both read heads — the circular settlement that reads density and the pulled band that reads boundary order — converge onto a shared plateau 2.8 to 3.7 percent from optimal, regardless of geometry. Two different local readers hitting one wall means the residual lives in a channel neither carries. This paper names the channel, proves why the two heads cannot see it, and reads it.

The proof is one line of Law 3. Attach any per-node field to the map — a price at every city. Over every closed tour, the field's contribution telescopes to a constant: the net field across a closed boundary is zero. Verified live to machine precision — maximum deviation 7.9×10^{-10} over 2,200 random closed tours across eleven instances. The consequence is structural, not algorithmic: any information carried by a node field is invisible to any reader that evaluates settled wholes, and both existing heads settle wholes. The shared plateau is not a performance failure. It is the signature of a Law-3-cancelling channel: the wall both heads hit is the wash.

The field that carries the residual is the closure constraint's own pricing. Law 3 demands the tour be one closed boundary — exactly two live pairings at every node — so a node's degree minus two is its local closure deficit, and the classical Held–Karp ascent (used here as a beam) is precisely the repricing loop the framework predicts: surcharge the over-paired, subsidize the under-paired, iterate until deficits clear. Theorem 2 of the minimal substrate said conservation cannot be enforced as one number and must be a gradient; the closure constraint obeys — its enforcement is a per-node field, computed live: the field's floor quote reaches the settled optimum exactly on berlin52 (7542.0, gap 0.000 percent) and sits within roughly one percent on most of the bench. The pairing-level readout of this field is Helsgaun's α — the cost to the whole of forcing one pairing into the minimal closure object — and it sees what the local channels cannot: on ch130 the optimal tour's deepest edge ranks 14th by distance and 2nd by α; at candidate depth eight, α covers 100 percent of optimal-tour edges on eight of nine tour-verified instances while raw distance misses on two. The probe: one identical crude engine, identical starts, only the

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candidate feed changed. The α feed wins the mean on 7 of 11 instances (sign test not significant, stated), but the magnitude concentrates exactly where the corpus said the local channels fail: pr152 — the settlement arm's recorded worst case — drops from 7.64 to 3.09 percent mean and 1.74 to 0.77 percent best, and pr76's exact optimum is found by the α feed alone; berlin52, the circular head's exact home, gains nothing, as it should. The third read head reads the ledger, not the map — and it must, because over any finished closure the ledger cancels.

1. The Plateau as a Measurement

From Nothing, Part V, closed on the open note that governs this strike. The circular read settles a slack field center-out and is exact where density is uniform; the linear read pulls the boundary edge-in and wins where density is structured; the min of both lands within 2.5 percent everywhere with no tuning — and then, at N = 100–150, both arms' margins collapse together into one band, 2.8 to 3.7 percent from optimal, on the most structured and the least structured instances alike. The report diagnosed it correctly and stopped: both are local-geometry readers — one reads density, one reads boundary order — and they appear to exhaust the same channel. The residual lives in a channel neither carries.

Treat the plateau as a measurement rather than a defeat, and it has told us three things about the missing channel before we look for it. It is shared — so it is not a defect of either head's mechanism. It is geometry-independent — so it is not a property of the point layout the heads read. And it is percent-scaled — so it grows with the object, like an accounting residue, not like a local error. What kind of channel is invisible to two different local readers at once? Section 2 answers with a theorem instead of a guess.

2. The Wash Theorem: Why No Whole-Tour Reader Can See It

Attach a field to the map: one number π_i at every city, any numbers at all. Reprice every pairing by the field, $c'(i,j) = c(i,j) + \pi_i + \pi_j$. Now walk any closed tour. Every city is entered once and left once — Law 3's closed boundary, exactly two live pairings per node — so every π_i is collected exactly twice, and the repriced tour length exceeds the true length by the constant $2\Sigma\pi$, for every closed tour, identically. The net contribution of a node field across a closed boundary is zero. This is not an approximation and not a property of good fields; it is bookkeeping, and it was verified live anyway, because the convention here is that even identities show their receipts: across eleven instances, 200 random closed tours each, the deviation of (repriced length − true length) from the constant never exceeded 7.9×10^{-10} — floating-point residue on 2,200 closures. FORCED as an identity; CONFIRMED at machine precision.

Now the consequence, which is the whole paper. If the residual channel is carried by a node field, then every reader that evaluates settled wholes — every reader whose verdict is a property of a finished closed tour — inherits the cancellation and reads nothing. The circular head settles a whole ring. The linear head pulls a whole boundary. Both are whole-tour readers. Both are therefore structurally blind to any node-field channel, and blind by the same mechanism, which is why they hit the same wall. The plateau is the wash: a channel whose total signal cancels over every closure, reading locally as absence. The framework has met this object before — at the bottom of everything. Total connection reads as no connection; a fully

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balanced ledger reads as nothing. The residual three percent is a balanced ledger folded into every candidate tour, and the only place it can be read is before the closure balances it: at the level of the open pairing.

3. The Field That Prices Closure

Which node field carries the residual? The one the constraint itself generates. Law 3's demand on a tour is local and countable: exactly two live pairings at every node. A node's pairing count minus two is its closure deficit — positive means over-subscribed, negative means starved. Theorem 2 of the minimal substrate proved that a constraint of this kind cannot be enforced by one global number; enforcement is forced into gradient form, a per-node flux. The construction that realizes this is classical and is used here as a beam, marked: Held and Karp, 1970. Build the cheapest object that contains exactly one closure — the 1-tree, a spanning tree plus one edge, the minimal Law-3 object. Read every node's deficit. Surcharge the over-paired, subsidize the starved — π_i moves with the deficit — rebuild the 1-tree in the repriced field, and iterate. The loop is the framework's repricing engine written thirty years before the framework: the constraint, prior to any object, generating the field that will select the object.

Two readouts of the ascent, both live. First, the floor quote: the repriced 1-tree's cost is a lower bound on every closed tour — the price of closure, quoted by the field before any tour is walked. A simple ascent (1,500 iterations, subgradient with momentum, no refinements) produced the floors below; they are floors of a weak negotiator, valid always, maximal never — pr76's 2.8 percent says the negotiator quit early there, not that the field is loose:

Instance	n	Optimum	Field floor	Gap %	Wash check: max deviation, 200 closures
berlin52	52	7542	7542.0	0.000	1.9e−11 — the quote equals the settled render
st70	70	675	671.0	0.594	2.4e−12
pr76	76	108159	105112.6	2.817	3.6e−10 (weak ascent, floor only)
rd100	100	7910	7898.6	0.144	3.9e−11
kroA100	100	21282	20936.1	1.625	1.3e−10
eil101	101	629	627.4	0.251	3.2e−12
lin105	105	14379	14370.3	0.061	9.0e−11
bier127	127	118282	117430.1	0.720	6.6e−10
ch130	130	6110	6075.0	0.573	4.7e−11
ch150	150	6528	6489.4	0.592	6.4e−11

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pr152	152	73682	72912.1	1.045	7.9e−10

Second, the location of the residual. On most of the bench the field's floor sits within roughly one percent of the optimum — and on berlin52 it is the optimum, to the digit. The local heads plateau near three percent; the field's quote sits near one. The missing channel spans exactly the space between the wall the whole-tour readers hit and the floor the field already knows — and by Section 2, that space is invisible from above the closure and readable only from below it.

4. The Veto: What the Closure Charges for a Pairing

The field cancels over wholes, so it must be read at the pairing level, and the pairing-level readout is the second beam, marked: Helsgaun's α, 2000. For a candidate pairing (i, j), α(i, j) is the increase in the minimal closure object's cost when that pairing is forced into it — the cost to the whole of accepting this edge, the closure's veto strength. Tree pairings carry α = 0: the closure already wanted them. A locally short edge the closure must be bent to accommodate carries a large α; a locally long edge the closure was already reaching for carries a small one. α is precisely the information the wash hides: it is defined before closure, on the open pairing, where the field has not yet cancelled.

Measured against the nine instances whose optimal tours are published (each tour re-verified to its known length under the TSPLIB metric before use), the channel content is direct. Rank every city's alternatives two ways — by raw distance, the local channel, and by α, the closure channel — and ask where the optimal tour's edges sit:

Instance	**Cov. k=5, dist**	**Cov. k=5, α**	**Worst rank, dist**	**Worst rank, α**	**Locally invisible edge (d-rank → α-rank)**
berlin52	100.0 %	100.0 %	5	2	—
st70	98.6 %	100.0 %	6	5	—
pr76	97.4 %	97.4 %	11	10	11 → 10 (deep in both; on record)
rd100	97.0 %	100.0 %	7	4	—
kroA100	99.0 %	98.0 %	13	8	13 → 8
eil101	100.0 %	100.0 %	3	4	—
lin105	96.2 %	100.0 %	7	5	—
ch130	96.2 %	99.2 %	14	6	14 → 2
ch150	98.7 %	100.0 %	7	4	—

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At candidate depth eight the separation is clean: α covers 100 percent of the optimal tour's edges on eight of the nine instances, while raw distance misses on kroA100 (99.0) and ch130 (99.2) — and the edges distance misses are exactly the ones α recovers. The ch130 row is the finding in miniature: the optimal tour contains a pairing that is the 14th choice of the local channel and the 2nd choice of the closure channel. That edge is the residual, exhibited — a pairing no local reader ranks and the field endorses. pr76 carries the one honest exception: its deepest optimal edge is deep in both channels (11th and 10th), and it stays on the record as the row that keeps this section honest.

5. The Probe: One Engine, Two Feeds

The channel measurement says the information exists; the probe asks whether reading it pays. Design: one deliberately crude improvement engine — 2-opt plus Or-opt over candidate lists of six, nearest-neighbor multistart, thirty identical start cities per instance — run twice per instance, changing exactly one thing: whether the candidate lists are ranked by distance or by α. The engine's absolute gaps are the engine's; the controlled variable is the feed.

Instance	Optimum	Dist: best	mean %	α: best	mean %	Δ mean (pts)
berlin52	7542	7542 (0.00)	4.98	7542 (0.00)	5.50	−0.51
st70	675	683 (1.19)	4.26	683 (1.19)	3.41	+0.85
pr76	108159	109067 (0.84)	5.04	108159 (0.00)	3.14	+1.89
rd100	7910	7916 (0.08)	4.55	7917 (0.09)	4.80	−0.25
kroA100	21282	21282 (0.00)	1.92	21379 (0.46)	2.12	−0.20
eil101	629	631 (0.32)	2.62	631 (0.32)	2.58	+0.04
lin105	14379	14529 (1.04)	10.11	14490 (0.77)	9.87	+0.24
bier127	118282	118696 (0.35)	3.51	118693 (0.35)	3.79	−0.27
ch130	6110	6228 (1.93)	4.54	6200 (1.47)	4.08	+0.45
ch150	6528	6553 (0.38)	2.33	6549 (0.32)	1.84	+0.49
pr152	73682	74967 (1.74)	7.64	74249 (0.77)	3.09	+4.56

Read the table the way the discipline demands, worst news first. The α feed loses the mean on four instances of eleven, by small margins (−0.20 to −0.51 points), and the 7-of-11 win count on its own is a coin flip dressed up — sign test $p \approx 0.27$, not significant, stated. The aggregate improvement, 4.68 to 4.02

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percent mean-of-means, is real but modest. If the claim were "α helps everywhere," the table would kill it.

That is not the claim. The claim is that the α channel carries what the local channels miss — and the magnitude column lands exactly on the corpus's own map of where the local channels fail. pr152 is the recorded worst case of the settlement arm — the instance where the √N law broke, the structured-density failure the whole dual-head construction was built around. Under the α feed it collapses: mean 7.64 to 3.09 percent, best tour 1.74 to 0.77 percent — through the dual-head plateau with a crude engine. pr76's exact optimum is reached by the α feed and not by the distance feed. And berlin52 — the circular head's exact home, the instance the local channel already renders perfectly — gains nothing and gives back half a point: where the local channel is already complete, the ledger has nothing to add, precisely as the wash predicts. The channel pays where the corpus said the wall was, and only there. The instance-family alignment is one family deep and is claimed as PULL, not proof; the pre-registered round that would promote it is specified in Section 6.

6. The Diagnosis, Assembled — and the Next Round, Pre-Committed

The plateau now has a mechanism with every link either forced or measured. The residual channel is carried by a per-node field (Section 3: the closure's own pricing, forced into gradient form by Theorem 2). A per-node field cancels over every closed boundary (Section 2: identity, verified to 7.9×10^{-10}). Both existing heads evaluate closed wholes. Therefore both are blind to the channel, therefore they share one wall. The third read head is not a better geometry reader; it is a reader of a different object — the open pairing, priced by the constraint before the closure washes the price out. The two local heads read the map. The third head reads the ledger. The handoff law gains its third veto: after anisotropy and density span, α.

One reframing is recorded as PULL because it is load-bearing for the program even though nothing below tests it directly. The ascent's field is dual, and the dual admits superposed, fractional pairings — half-edges the LP relaxation allows and no rendered tour can contain. In the corpus's channel language: the field lives in the asking half, the settled tour in the answering half, and the gap between the field's floor and the rendered optimum is the collapse cost — the price of forcing a superposed pairing structure into one closed readout. If that reading is right, the gap between floor and optimum is not slack to be engineered away but a measurement of the render's price, and it should behave like one: instance-stable, geometry-linked, and never negative. OPEN to test.

The next round, pre-committed before any of it runs. One: the blind handoff round 3 adds the α veto to the two standing vetoes, with winners and decisiveness margins pre-registered — the handoff law reaches significance there or dies there, as already promised. Two: the α channel is coupled into the settlement arm itself — not by repricing settled rings, which the wash forbids and Section 2 now proves pointless, but by feeding the elastic couplings at the pairing level, K(x) from local density and coupling admission by α. Three: scale — the plateau was measured at N = 100–150; the claim that the α channel breaks it must be tested there and above with matched engines, α feed against distance feed, margins declared

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decidable in advance. If the α feed fails to separate on the pre-registered structured family, the third-head identification is wrong and the ledger will say so.

7. Status Ledger

Claim	Status	Basis
Node-field contribution cancels over every closed tour	FORCED + CONFIRMED	Telescoping identity; max deviation 7.9e−10 over 2,200 random closures, 11 instances
Whole-tour readers are blind to node-field channels	FORCED	Direct consequence of the cancellation
Shared plateau of the two local heads caused by a Law-3-cancelling channel	PULL (strong)	Mechanism forced; attribution to the residual pending round 3
Closure enforcement is a per-node gradient (deg − 2 as local deficit)	FORCED (in-frame) + BEAM	Theorem 2 pattern; Held–Karp 1970 as the realizing construction
Field floor quotes, 11 instances	CONFIRMED (floors)	0.000 % (berlin52, exact) to 2.817 % (pr76, weak ascent); valid lower bounds always
α as pairing-level readout of the closure field	BEAM + LIVE	Helsgaun 2000; computed from best-ascent 1-tree
α channel carries locally invisible optimal pairings	CONFIRMED	ch130: d-rank 14 → α-rank 2; α covers 100 % at k = 8 on 8/9; distance misses 2/9
pr76 deep edge deep in both channels	ON RECORD	Ranks 11 and 10; the honest exception
α feed breaks the recorded failure case	CONFIRMED	pr152: mean 7.64 → 3.09 %, best 1.74 → 0.77 %; pr76 optimum found by α feed only
α feed helps everywhere	NOT SUPPORTED	4/11 small negative deltas; sign test p ≈ 0.27; aggregate +0.66 pts only
Field adds nothing where local channel already exact	CONFIRMED	berlin52: −0.51 pts; consistent with the wash
Dual field as asking-channel object; floor-to-optimum gap as collapse cost	PULL	Structural mapping; untested
α veto joins the handoff law; settlement-arm coupling; scale round	OPEN / PRE-COMMITTED	Round 3 blind, margins pre-registered

8. Closing: The Ledger Reads Back

The tension-fall solver died because it consumed the closure one mark at a time and lost the whole; its epitaph was that the mark carries the instruction only when the mark is the whole chain. This strike is the same lesson from the other side. The whole chain, once closed, hides a channel — the field balances over

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every closure and reads as absence, the wash doing at the scale of a tour what it does at the scale of everything. So the residual was never going to be read by a better settler or a sharper band: both hold finished wholes, and the information dies in the closing. It lives only in the open pairing, where the constraint has priced the map and nothing has yet cancelled the price.

Constraint prior to the object, one more time and literally: the field exists before any tour, quotes a floor no tour can beat, endorses pairings no local reader would rank, and on the bench's recorded failure case that endorsement cut the wall from seven percent to three with nothing changed but the feed. The two heads read the map. The third head reads what the map owes.

Appendix: Live Output, Quoted Verbatim

Selected lines, unedited, from the three fixed programs of this session (hk.py, solver.py):

```
inst         n     opt optfile     HK-LB   gap%  wash(max dev)
berlin52    52    7542    7542    7542.0  0.000  1.89e-11
pr76        76  108159  108159  105112.6  2.817  3.59e-10
lin105     105   14379   14379   14370.3  0.061  9.00e-11
pr152      152   73682      --   72912.1  1.045  7.86e-10


ch130      5   96.2   99.2   worst d-rank 14   worst a-rank 6   invisible: [(14, 2)]
kroA100    5   99.0   98.0   worst d-rank 13   worst a-rank 8   invisible: [(13, 8)]
pr76       5   97.4   97.4   worst d-rank 11   worst a-rank 10  invisible: [(11, 10)]


pr76      108159 |    109067  0.84  5.04 |    108159  0.00  3.14 | +1.89
pr152      73682 |     74967  1.74  7.64 |     74249  0.77  3.09 | +4.56
berlin52    7542 |      7542  0.00  4.98 |      7542  0.00  5.50 | -0.51
mean-of-means gap: dist-feed 4.68%  alpha-feed 4.02%  (alpha better on 7/11)
```

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